

Linear combinations of linearly independent vectors

- ▶ suppose x is linear combination of linearly independent vectors $a_1, \dots, a_k$:

$$x = \beta_1 a_1 + \dots + \beta_k a_k$$

- ▶ the coefficients $\beta_1, \dots, \beta_k$ are *unique*, i.e., if

$$x = \gamma_1 a_1 + \dots + \gamma_k a_k$$

then $\beta_i = \gamma_i$ for $i = 1, \dots, k$

- ▶ this means that (in principle) we can deduce the coefficients from x
- ▶ to see why, note that

$$(\beta_1 - \gamma_1)a_1 + \dots + (\beta_k - \gamma_k)a_k = 0$$

and so (by linear independence) $\beta_1 - \gamma_1 = \dots = \beta_k - \gamma_k = 0$

Outline

Linear independence

Basis

Orthonormal vectors

Gram–Schmidt algorithm

Independence-dimension inequality

- ▶ *a linearly independent set of n -vectors can have at most n elements*
- ▶ *put another way: any set of $n + 1$ or more n -vectors is linearly dependent*

Basis

- ▶ a set of n linearly independent n -vectors $a_1, \dots, a_n$ is called a *basis*
- ▶ any n -vector b can be expressed as a linear combination of them:

$$b = \beta_1 a_1 + \dots + \beta_n a_n$$

for some $\beta_1, \dots, \beta_n$

- ▶ and these coefficients are unique
- ▶ formula above is called *expansion of b in the $a_1, \dots, a_n$ basis*
- ▶ example: $e_1, \dots, e_n$ is a basis, expansion of b is

$$b = b_1 e_1 + \dots + b_n e_n$$

Outline

Linear independence

Basis

Orthonormal vectors

Gram–Schmidt algorithm

Orthonormal vectors

- ▶ set of n -vectors $a_1, \dots, a_k$ are (*mutually*) *orthogonal* if $a_i \perp a_j$ for $i \neq j$
- ▶ they are *normalized* if $\|a_i\| = 1$ for $i = 1, \dots, k$
- ▶ they are *orthonormal* if both hold
- ▶ can be expressed using inner products as

$$a_i^T a_j = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

- ▶ orthonormal sets of vectors are linearly independent
- ▶ by independence-dimension inequality, must have $k \leq n$
- ▶ when $k = n$, $a_1, \dots, a_n$ are an *orthonormal basis*

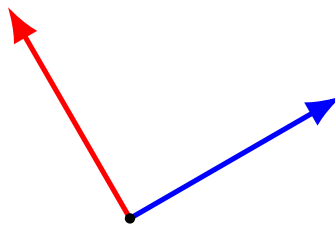
Examples of orthonormal bases

- ▶ standard unit n -vectors $e_1, \dots, e_n$

- ▶ the 3-vectors

$$\begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}, \quad \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

- ▶ the 2-vectors shown below



Orthonormal expansion

- ▶ if $a_1, \dots, a_n$ is an orthonormal basis, we have for any n -vector x

$$x = (a_1^T x)a_1 + \dots + (a_n^T x)a_n$$

- ▶ called *orthonormal expansion of x* (in the orthonormal basis)
- ▶ to verify formula, take inner product of both sides with a_i